



AntiScam AI

Real-Time Digital Fraud Interception Platform

Intercepting conversational fraud — before the money moves.

ET AI Hackathon 2026

Problem 6 — AI for Digital Public Safety: Defeating Counterfeiting, Fraud & Digital Arrest Scams

github.com/Pavankumar06T/AntiScam-AI

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1 Executive Summary

AntiScam AI is a multi-agent AI platform that detects social-engineering scams — "digital arrest", KYC, lottery, loan, job and investment fraud — from a live call or chat transcript, and intervenes *while the call is still happening*.

It scores scam risk continuously as a conversation unfolds, cross-references the caller's identifiers against a fraud-network graph to reveal a repeat scammer already working other victims, delivers a plain-language warning in the victim's language, and — on confirmation — auto-drafts a structured police complaint and a dispatch-ready containment order to freeze the mule account.

The core insight driving the project: **law enforcement lacks intelligence *before* mass victimisation, not after**. Existing tools are forensic — they help once the complaint is filed and the money is gone. AntiScam AI operates at the point of contact, shifting fraud response from reactive investigation to proactive interception, and — critically — from treating each victim in isolation to mapping the organised network behind them.

~2s

DETECTION LATENCY

0%

FALSE-POSITIVE RATE*

84

AUTOMATED TESTS

3

LANGUAGES (EN/HI/TA)

* On the hard-negative subset — legitimate calls engineered to look scam-like. See §12. This document describes each component of the system in detail, followed by the architecture, technology, results, and design rationale.

2 The Problem

India recorded **1.14 million cybercrime complaints in 2023**. "Digital arrest" scams — where fraudsters impersonate CBI, ED or Customs officers over a video call and psychologically hold victims hostage until they transfer money — defrauded citizens of **over ₹1,776 crore in just the first nine months of 2024**.

How the digital-arrest playbook works

The scam is not random; it follows a disciplined psychological script. A caller establishes a **pretext** (an intercepted parcel, a money-laundering case), **asserts authority** (a badge number, an FIR reference), **induces fear** (threats of arrest, a non-bailable warrant), **isolates** the victim (forbidding them from telling family, invoking the "Official Secrets Act"), and finally **extracts** money to a "safe" or "RBI verification" account. Victims comply because the pressure is relentless, the story is internally consistent, and they are cut off from anyone who could break the spell.

These are not opportunistic crimes. They are **industrialised operations** run from fraud compounds, using spoofed numbers, AI-generated voices, fake government portals, and networks of mule bank accounts that move money within minutes — often across several victims and several cities at once.

Why current tools fall short

- **They act after the fact.** Complaint portals and transaction-fraud systems engage once money has already moved. By then the funds have been layered through mule accounts and are effectively unrecoverable.
- **They judge one conversation in isolation.** They cannot see that the same account or phone number is simultaneously working dozens of other victims — so each case is investigated as if it were unique, and the operation behind it is never mapped.
- **They don't reach the victim in the moment** — during the call, when a 30-second, regulation-backed warning could stop the transfer entirely.

The gap AntiScam AI fills

Intelligence **before** victimisation — detecting the scam from language and dialogue in real time, linking scammers across victims into one network, and warning the target before the extraction step.

3 Our Solution — Overview

AntiScam AI is a **four-agent system** orchestrated as a state machine. A transcript chunk flows through detection, and only if risk crosses a threshold does it fan out to the network, advisory and reporting agents — so ordinary calls cost almost nothing while genuine scams trigger the full response. Each agent is described in detail in the sections that follow.

Agent	Role	Core tech
Scam Pattern Detection	Scores scam risk (0–100) from language and dialogue dynamics; classifies scam type, red flags, escalation stage, and extracts scammer identifiers.	Rules + Groq LLM
Fraud Network Graph	Cross-references a session's identifiers against every prior one to detect a repeat scammer and surface the wider operation.	NetworkX
Advisory (RAG)	Retrieves relevant guidance and generates a plain-language, cited warning in English, Hindi or Tamil.	ChromaDB + LLM
Evidence & Reporting	Assembles an NCRB-style complaint draft and a dispatch-ready disruption packet from confirmed-fraud sessions.	Deterministic + LLM

The four map onto the verbs of the challenge statement: **detect** (detection), **link** and **disrupt** (graph + reporting), and **respond** (advisory). The orchestrator is a LangGraph graph with conditional edges, so the routing itself is part of the design rather than hard-coded control flow.

4 Detection Methodology

The Detection Agent is the heart of the system, and it is deliberately **two layers, fused** rather than a single model call.

Layer 1 — deterministic rule tripwires

A curated lexicon of named coercion patterns is matched against the transcript in microseconds: authority impersonation, the "digital arrest" phrase, arrest threats, fabricated case references, isolation instructions, secrecy demands, the "safe/verification account" transfer, OTP/PIN requests, remote-access app installs, verification pretexts, advance fees, unrealistic rewards, guaranteed-return lures, and fast payment rails. Patterns include common Hindi/Hinglish transliterations, since real scam calls code-switch constantly. Each hit carries a weight; the weights are summed and passed through a **saturating curve** so that stacking many weak signals approaches — but never reaches — certainty, while two or three critical signals get most of the way there. No single unlucky keyword can max out the score.

Layer 2 — the Groq LLM judgement

The transcript is sent to `llama-3.3-70b-versatile` via Groq with a carefully engineered system prompt, four few-shot examples (including a legitimate call that *looks* alarming, as a false-positive control), JSON-mode output, a low temperature for consistency, and a fixed seed for reproducible scoring. The model returns the scam type, red flags with verbatim quotes, an escalation stage, a confidence, and reasoning.

Fusion, escalation stage, and identifiers

The two scores are fused as a weighted blend (LLM-dominant, ~75/25). The system also reports the highest **escalation stage** reached on the coercion script — no contact risk → pretext → authority → fear → isolation → extraction — which is what makes it possible to warn *before* the extraction step and to measure interception lead time. Scammer identifiers (phone, UPI, account, case number, URL, claimed name and department) are extracted **deterministically by regex**, never by the LLM, because a hallucinated account number would poison the fraud graph and any evidence packet.

Why fuse two layers?

It buys four things: a **latency floor** (the UI moves before the LLM round-trip lands); **graceful degradation** (if Groq is rate-limited or down, an obvious scam still scores in the 90s on rules alone); **explainability** ("it matched these named patterns" is auditable in a way "the model said 82" is not); and **drift protection** (a prompt edit cannot silently erase a hard signal like "share your OTP").

5 Fraud Network Graph

This agent is the project's key differentiator. A detector that reads one conversation can only ever say "this looks like a scam." A graph that remembers every prior flagged session can say something far more actionable: *"the account this caller wants you to pay was already used against two other people this week."*

How it is built

Sessions and identifiers live in one **bipartite graph** (NetworkX): each session links to the identifiers seen in it, so two sessions sharing an identifier sit at distance two, and a connected component *is* a fraud operation spanning multiple victims. When a new session arrives it is cross-referenced against the graph *before* being ingested, so it can never match itself.

What it links on — and what it deliberately does not

Linking is restricted to identifiers that genuinely implicate a shared operator — phone, UPI, bank account, case number, URL — each with a strength weight (a reused mule account or UPI is near-conclusive; a reused case number is weaker). It **never links on claimed name or department**: thousands of unrelated scammers all say "CBI" and "Inspector Sharma", so linking on those would collapse the entire dataset into one meaningless cluster and risk naming innocent people. That guard is enforced by automated tests.

The victim count reports only *other* victims (excluding the current caller, who is being protected rather than counted), so the number is stable and honest — "this scammer has already hit 3 other people" — regardless of whether the live session has been ingested yet.

6 Advisory Agent (RAG)

The Advisory Agent is the only component the victim actually reads; everything else is instrumentation. When risk crosses the threshold it retrieves relevant guidance and generates a warning addressed directly to the person on the call.

Retrieval that costs nothing

A curated knowledge base of MHA/RBI-style scam advisories and statutory references is embedded in **ChromaDB using local ONNX MiniLM embeddings** — so retrieval runs entirely offline, costs zero API tokens, and stays available even when the LLM layer is rate-limited. Retrieval is semantic but biased toward advisories tagged for the detected scam type, so a digital-arrest victim reliably gets the "digital arrest is not real" advisory.

Multilingual, and resilient

The warning is generated in **English, Hindi or Tamil** with a decisive headline, a short plain-language body, numbered immediate actions, and a citation to the advisory it drew on. If the LLM is unavailable, the agent falls back to **hand-written templates per language** — never machine-translated, because a mistranslated safety instruction is worse than none. The warning gets blunter, not absent, precisely when the person needs it most.

7 Evidence, Reporting & Disruption

Once a victim confirms the call was fraud, two artefacts are assembled.

The NCRB-style complaint draft

A victim files a complaint hours later, from memory, while distressed — and the details that let police act (the exact UPI ID, the account number, the precise words used, the timestamps) are exactly the details that evaporate under stress. The Reporting Agent captured them while the call was happening. The draft includes a chronological incident summary, suspect identifiers, **timestamped red-flag excerpts**, the fraud-cluster reference, illustrative statutory provisions (BNS 2023 / IT Act), and where to file (1930 / cybercrime.gov.in). Construction is mostly deterministic — only the prose summary is written by the LLM — so an invented account number can never end up in a police document.

The disruption packet — the "disrupt" step

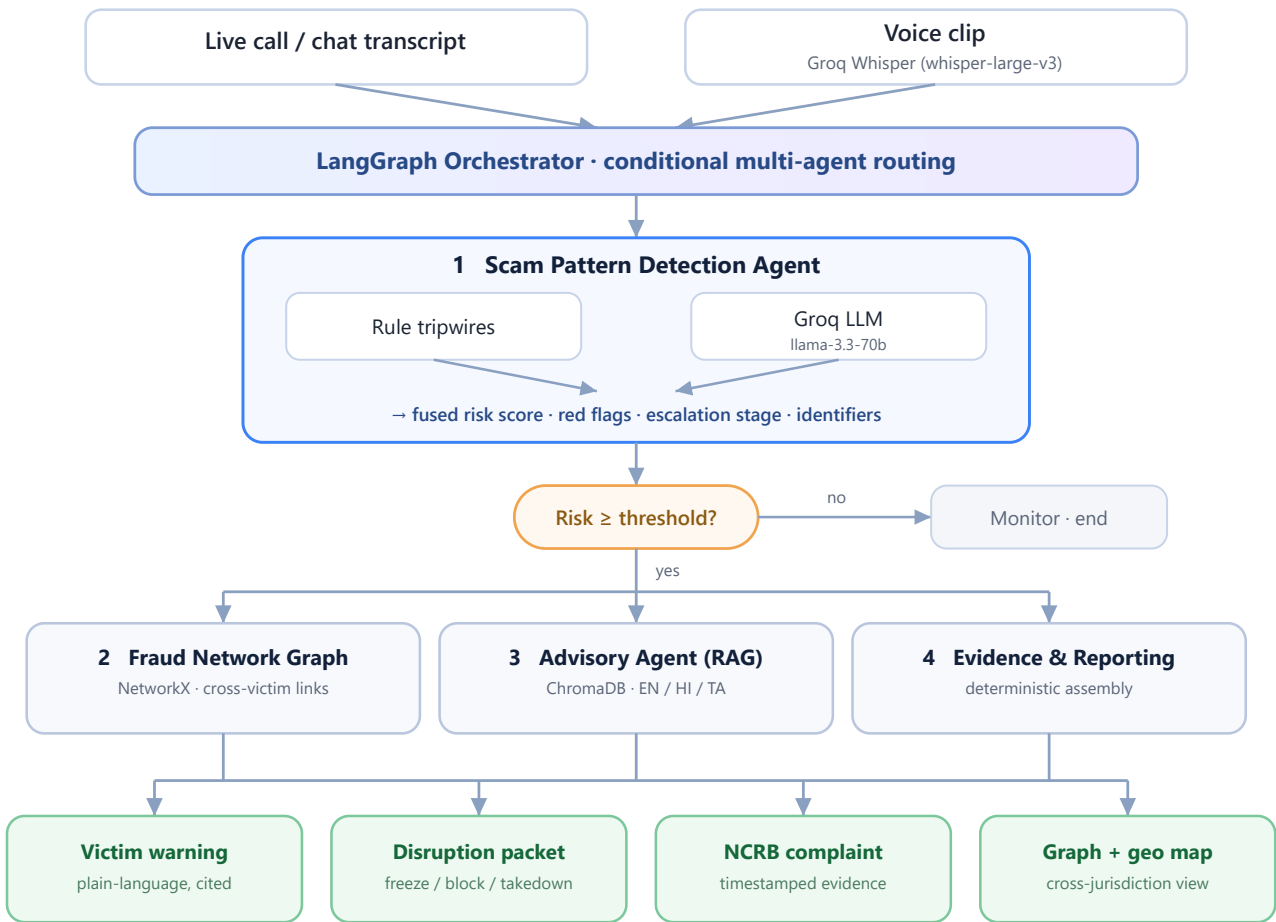
Detection protects the person on this call; the disruption packet proposes cutting the operation off at the infrastructure. It is a dispatch-ready containment order — **freeze the mule account, block the caller's number, take down the phishing URL** — routed to the beneficiary bank / NPCI, the telecom operator / DoT (Sanchar Saathi), and CERT-In. Because the identifier is shared across a cluster, actioning it protects *every linked victim at once*. It is assembled deterministically from captured identifiers, carries a package id and timestamp, and is court-auditable.

Prototype boundary — stated honestly

In this prototype the disruption packet is **generated, not auto-dispatched**: a human operator would forward it to the bank/1930. In production it would be sent automatically via bank/NPCI and DoT APIs. The value demonstrated here is the intelligence packaging and auditability.

8 System Architecture

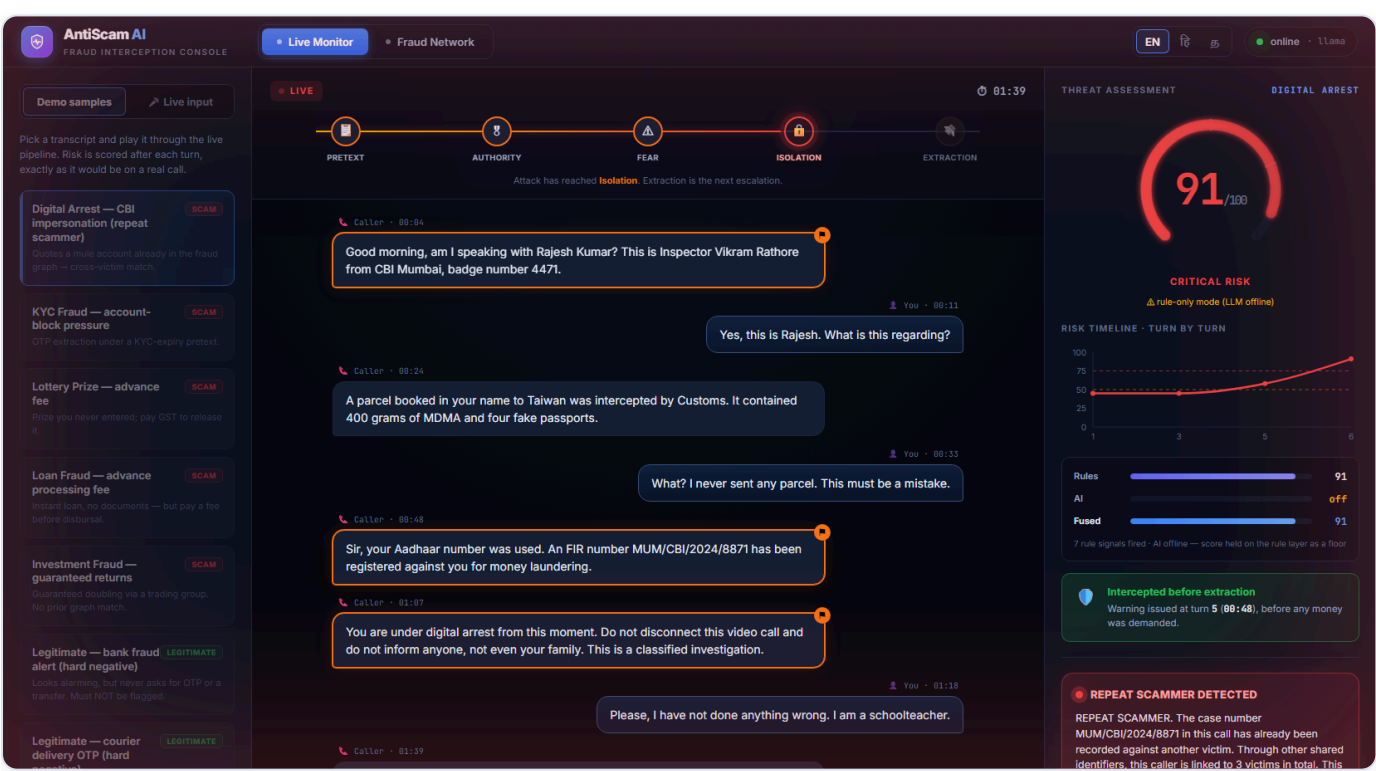
A LangGraph orchestrator routes each transcript through the agents with conditional branching — the graph, advisory and reporting agents only run once risk crosses the warning threshold. Live voice is transcribed by Groq Whisper before entering the same pipeline.



9 Technology Stack

Layer	Technologies & rationale
Backend	Python 3.12, FastAPI, Pydantic — typed schemas as the contract between agents, API and UI
Orchestration	LangGraph (multi-agent state machine) + LangChain — conditional routing is part of the design, not hard-coded control flow
LLM	Groq API — llama-3.3-70b-versatile (detection, advisory, reporting) chosen for speed; whisper-large-v3 (voice)
Retrieval (RAG)	ChromaDB with local ONNX MiniLM embeddings — runs offline, zero token cost, resilient to rate limits
Fraud graph	NetworkX — in-memory bipartite session/identifier graph; connected components give clusters for free
Frontend	React 19, Vite, Recharts (risk timeline), react-force-graph-2d (network), custom SVG (geo map)
Quality	84 automated tests (pytest); a colorblind-validated data-visualisation palette

10 The System in Action



Live Session Monitor. The digital-arrest call playing turn-by-turn: the coercion kill-chain lit to Isolation, the risk gauge at 91 (Critical), the rule/AI/fused breakdown, and the interception callout confirming the warning fired before any money was demanded.

AntiScam AI

FRAUD INTERCEPTION CONSOLE

Live Monitor

Fraud Network

EN

🔍

📄

🟢 online · 11:55

Demo samples

Live input

Pick a transcript and play it through the live pipeline. Risk is scored after each turn, exactly as it would be on a real call.

Digital Arrest — CBI impersonation (repeat scammer)

Quoted a mule account already in the fraud graph → cross-victim match.

KYC Fraud — account-block pressure

OTP extraction under a KYC-expiry pretext.

Lottery Prize — advance fee

Prize you never entered, pay GST to release it.

Loan Fraud — advance processing fee

Instant loan, no documents — but pay a fee before disbursal.

Investment Fraud — guaranteed returns

Guaranteed doubling via a trading group. No prior graph match.

Legitimate — bank fraud alert (hard negative)

Looks alarming, but never asks for OTP or a transfer. Must NOT be flagged.

Legitimate — courier delivery OTP (hard negative)

LIVE

01:52

PRETEXT

AUTHORITY

FEAR

ISOLATION

EXTRACTION

Attack at extraction — this is the moment money would be lost.

A parcel booked in your name to Taiwan was intercepted by Customs. It contained 400 grams of MDMA and four fake passports.

You · 00:33

What? I never sent any parcel. This must be a mistake.

Caller · 00:48

Sir, your Aadhaar number was used. An FIR number MUM/CBI/2024/8871 has been registered against you for money laundering.

Caller · 01:07

You are under digital arrest from this moment. Do not disconnect this video call and do not inform anyone, not even your family. This is a classified investigation.

You · 01:18

Please, I have not done anything wrong. I am a schoolteacher.

Caller · 01:39

To prove your funds are clean, you must transfer your balance to the RBI verification account 50100294471882. It will be refunded within two hours after verification.

Caller · 01:52

If you delay, a non-bailable warrant will be issued and you will be arrested tonight. Do it now.

Analysing conversation...

DO NOT transfer any money, and do not share any OTP or PIN.

Tell a family member or friend what just happened, right now.

Call the National Cyber Crime Helpline on 1930.

If you already paid, call 1930 immediately — the first hour matters most.

Grounded in: ADV-DA-001 ADV-LAR-001 ADV-DA-002

DISRUPT · DISPATCH TO BANKS & TELECOM

Disruption Package

DISRUPT-20260719-EPEAT - protects 3 linked victims

FREEZE

BANK ACCOUNT

50100294471882

→ 3 VICTIMS

→ Beneficiary bank fraud desk / NPCI

This caller is linked to 3 victims (cluster CLUSTER-001). Actioning the shared identifiers below disrupts the operation for every linked victim, not just this one. Dispatch immediately — mule funds move within minutes.

Copy dispatch packet

RED FLAGS 7

AUTHORITY IMPERSONATION

HIGH

"Good morning, am I speaking with Rajesh Kumar? This is Inspector Vikram Rathore from CBI Mumbai, badge number 4471."

Caller claims to be a law-enforcement or regulatory officer. Indian agencies do not conduct investigations or demand money over phone or video calls.

AUTHORITY IMPERSONATION

MEDIUM

ECI: your Aadhaar number has been used to verify your...

Warning & Disruption. The victim-facing warning with numbered actions grounded in cited advisories, and the dispatch-ready Disruption Package — freeze the mule account (linked to 3 victims), routed to the bank fraud desk / NPCI.

AntiScam AI

FRAUD INTERCEPTION CONSOLE

Live Monitor

Fraud Network

EN

🔍

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🟢 online · 11:55

Fraud Network Intelligence

Nodes shared between victim sessions are cross-victim links — the fingerprint of one organised operation.

Network graph

Geographic map

SEED-003

9876501234

Vikram Rathore

SEED-001

MUM/CBI/2024/8871

LIVE-digital_arr_

CBI

50100294471882

RBI

SEED-002

SEED-004

kbcclain2024@ybt

Rahul Verma

SEED-005

NODE TYPES

Victim session

HUB

Phone number

LINKS

UPI ID

LINKS

Bank account

LINKS

Case number

LINKS

Website

LINKS

Claimed name

Claimed agency

NETWORK OVERVIEW

7 VICTIM SESSIONS

3 OPERATIONS

20 IDENTIFIERS

32 LINKS

Reset to seed data

Click a node to inspect it. The red, glowing nodes are identifiers shared between multiple victims — the cross-victim links that expose an organised operation rather than isolated incidents.

Fraud Network — Graph. Victim sessions (hubs) linked through shared identifiers. The glowing red nodes are mule accounts and numbers reused across victims — the cross-victim links that expose one organised operation.



Fraud Network — Geographic Map. Victims plotted by city with cross-victim arcs. One digital-arrest operation is working Mumbai, New Delhi and Bengaluru with the same identifiers — a coordinated, cross-jurisdiction campaign, not isolated local incidents.

11 Alignment with the Problem Statement

Suggested build areas

Area	Coverage
Digital Arrest Scam Detection & Alerting	Core — deep
Fraud Network Graph Intelligence	Core — deep
Geospatial Crime Pattern Intelligence	Yes — command-centre map
Citizen Fraud Shield (multi-channel advisory)	Yes — EN/HI/TA warnings + guided reporting
Speech AI (voice input)	Yes — Groq Whisper transcription
Counterfeit Currency Identification	Out of scope — a separate computer-vision track

Evaluation focus

Judges measure	How AntiScam AI delivers
Digital-arrest detection precision & recall	Fused rule+LLM detector; metrics harness reports both
Fraud-network lead time before mass victimisation	The interception lead-time is a headline feature
Very low false-positive rate for citizen tools	0% on hard-negative legitimate calls (\$12)
Auditability for legal admissibility	Timestamped evidence + deterministic identifiers + complaint packet

Judging criteria

Criterion	Weight	Our strength
Innovation	25%	Cross-victim graph, kill-chain, lead-time, disruption packets
Business Impact	25%	Directly addresses a ₹1,776 cr problem; protects networks, not just individuals
Technical Excellence	20%	Multi-agent orchestration, fused detector, 84 tests, honest degradation
Scalability	15%	Stateless agents, conditional routing, deploy-ready configs
User Experience	15%	Real-time console; plain-language, multilingual victim warnings

12 Performance & Metrics

Metric	Result
Detection latency (LLM path)	1.8 – 2.5 seconds
Obvious scam score	90+ / 100
Legitimate-call score	< 10 / 100
Precision (rule-layer floor)	1.0
False-positive rate on hard negatives	0.0
Automated tests passing	84 / 84
Degraded-mode resilience	Full pipeline runs with the LLM offline

Evaluation uses a hand-curated benchmark that deliberately includes **hard negatives** — legitimate calls engineered to look scam-like (a genuine bank fraud alert, a real delivery OTP, an overdue-EMI reminder, a recruiter). A detector that only handles easy negatives would report a flattering false-positive rate that means nothing; these are the cases that matter for a citizen-facing tool. The rule-layer floor is verified with zero external calls, so the numbers are reproducible by anyone; full rule+LLM figures are generated by the metrics harness when a Groq key is configured.

13 Design Decisions & Engineering Discipline

Several deliberate choices distinguish this from a demo that merely calls an LLM:

- **Fail fast, don't stall.** The Groq client is built with `max_retries=0`. The SDK's default retry silently sleeps ~25 seconds inside a rate-limited call; in a real-time interception system a silent stall is worse than an honest rule-only score, so the system degrades to the rule layer in under 500 ms and says so.
- **Deterministic identifiers.** Everything that becomes a graph node or a line in a police complaint is extracted by regex, never by the model — because a hallucinated account number would be actively harmful and would break cross-victim linking.
- **One source of truth for thresholds.** The warn/urgent thresholds live in configuration, shared by the API, the orchestrator and the dashboard, so they can never disagree about what "warning" means.
- **No secrets in code.** All keys load from the environment; `.env` is gitignored and no key is hardcoded anywhere in the repository's history.

- **Accessible visualisation.** The fraud-graph node palette was validated for colour-blindness rather than eyeballed, and risk is always shown with a text label, never colour alone.
- **84 automated tests** cover the rule layer, the fusion and degradation logic, the graph's linking and false-linking guards, the orchestrator's routing, and the API contract.

14 Running the Project

Backend

```
cd backend · python -m venv venv · venv\Scripts\activate · pip install -r requirements.txt · create backend/.env with your GROQ_API_KEY · uvicorn app.main:app --reload --port 8080
```

Frontend

```
cd frontend · npm install · npm run dev · open http://localhost:5173
```

A Groq API key is free at console.groq.com. Without a key the system still runs in degraded (rule-only) mode, so the dashboard is demoable with zero quota.

Suggested demo path

1. Play the **Digital Arrest** sample and watch the risk climb, the kill-chain light up, and the repeat-scammer match appear.
2. Confirm fraud to generate the complaint draft and the disruption packet.
3. Switch to **Fraud Network** → **Geographic map** to see the operation spanning cities.
4. Use **Live input** to speak a scam line and watch it transcribe and score in real time.

15 Limitations & Future Work

Honest limitations

- **Synthetic data.** Transcripts and seed sessions are machine-generated and hand-authored; metrics reflect behaviour on this data, which is cleaner than reality.
- **No live telecom integration.** Calls are replayed or spoken into the demo, not tapped from a live line — the hard problem is systems integration, not the AI.
- **Regulatory/legal content is illustrative.** Advisory text and BNS/IT Act references are representative for a prototype, not verified legal advice.
- **In-memory fraud graph.** Production needs a persistent graph database and a legal basis for retaining this data.
- **English-dominant** detection, with common Hindi/Hinglish patterns covered.

Roadmap to production

- On-device or telecom-level speech-to-text for live call interception.
- Automated dispatch of disruption packets to bank/NPCI and DoT APIs.
- Persistent graph store (Neo4j) with privacy and retention controls.
- Expansion to all 12 regional languages and additional scam typologies.
- WhatsApp / IVR channels for citizen-initiated checks.
- A learned rule layer to complement the hand-built lexicon.